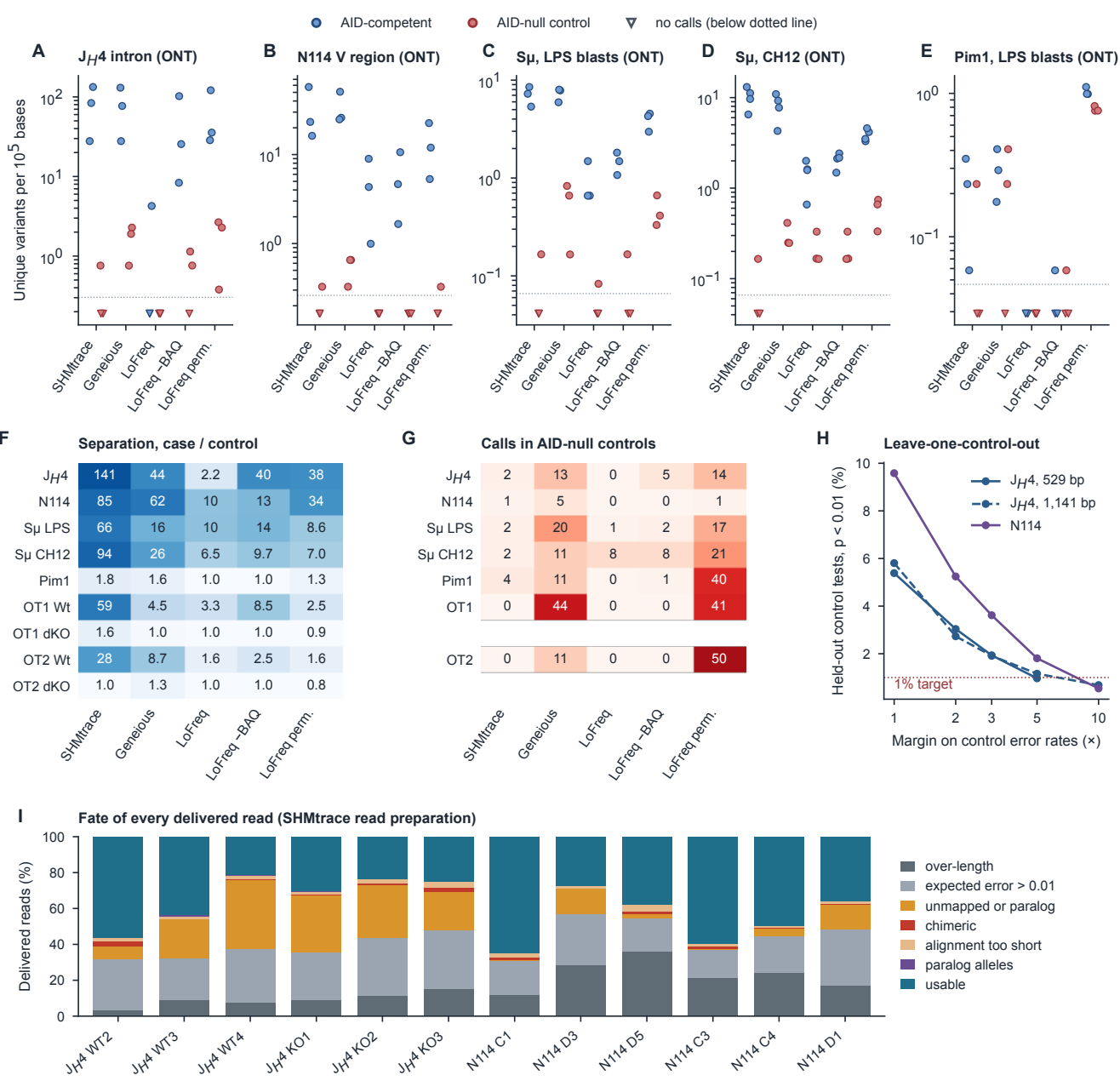

AID-competent
AID-null control
no calls (below dotted line)
A
J_H4 intron (ONT)
B
N114 V region (ONT)
C
Sμ, LPS blasts (ONT)
D
Sμ, CH12 (ONT)
E
Pim1, LPS blasts (ONT)
Unique variants per 10^5 bases
SHMtrace
Geneious
LoFreq
LoFreq −BAQ
LoFreq perm.
F
Separation, case / control
J_H4 141 44 2.2 40 38
N114 85 62 10 13 34
Sμ LPS 66 16 10 14 8.6
Sμ CH12 94 26 6.5 9.7 7.0
Pim1 1.8 1.6 1.0 1.0 1.3
OT1 Wt 59 4.5 3.3 8.5 2.5
OT1 dKO 1.6 1.0 1.0 1.0 0.9
OT2 Wt 28 8.7 1.6 2.5 1.6
OT2 dKO 1.0 1.3 1.0 1.0 0.8
G
Calls in AID-null controls
J_H4 2 13 0 5 14
N114 1 5 0 0 1
Sμ LPS 2 20 1 2 17
Sμ CH12 2 11 8 8 21
Pim1 4 11 0 1 40
OT1 0 44 0 0 41
OT2 0 11 0 0 50
H
Leave-one-control-out
Held-out control tests, p < 0.01 (%)
J_H4, 529 bp
J_H4, 1,141 bp
N114
1% target
Margin on control error rates (×)
I
Fate of every delivered read (SHMtrace read preparation)
Delivered reads (%)
over-length
expected error > 0.01
unmapped or paralog
chimeric
alignment too short
paralog alleles
usable
J_H4 WT2
J_H4 WT3
J_H4 WT4
J_H4 KO1
J_H4 KO2
J_H4 KO3
N114 C1
N114 D3
N114 D5
N114 C3
N114 C4
N114 D1